

Multiclass Classification Networks

Softmax Regression, Cross-Entropy, and Linear Decision Boundaries

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Roadmap

- ➊ Multiclass softmax model
- ➋ One-hot target representation
- ➌ Categorical cross-entropy learning objective
- ➍ Softmax Jacobian and gradient derivation
- ➎ Gradient-descent parameter updates
- ➏ Classification decision and linear boundaries
- ➐ Three-class numerical/Python example
- ➑ Assignment: handwritten-digit recognition

Multiclass Classification

- Multiclass classification assigns an input sample to one of $K \geq 3$ classes.
- We introduce the probabilistic **softmax** (maximum-entropy) classifier.
- Inputs may be float, binary, or integer features.
- Outputs are categorical class labels represented using one-hot vectors.

Goal

Learn parameters **W** and **b** so that the predicted class probabilities match the training targets.

One-Hot Representation

For $K = 3$ classes, a target vector contains exactly one 1:

$$A = [1, 0, 0], \quad B = [0, 1, 0], \quad C = [0, 0, 1].$$

- Each vector identifies the target class.
- The components sum to one.
- This representation is directly compatible with the softmax probability vector.

The Multiclass Network

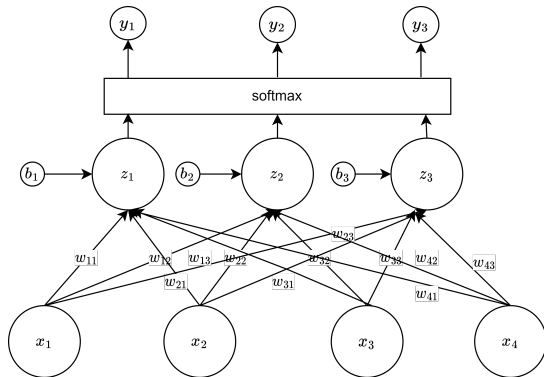


Figure: Four input variables and three output classes.

Linear Multiclass Model

For four inputs and three classes:

$$z_1 = w_{11}x_1 + w_{12}x_2 + w_{13}x_3 + w_{14}x_4 + b_1,$$

$$z_2 = w_{21}x_1 + w_{22}x_2 + w_{23}x_3 + w_{24}x_4 + b_2,$$

$$z_3 = w_{31}x_1 + w_{32}x_2 + w_{33}x_3 + w_{34}x_4 + b_3.$$

Compactly,

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$$

where $\mathbf{W} \in \mathbb{R}^{K \times d}$ and $\mathbf{b} \in \mathbb{R}^K$.

Softmax Output

The logits are transformed into normalized class probabilities:

$$y_i = \frac{e^{z_i}}{\sum_{c=1}^K e^{z_c}}, \quad i = 1, \dots, K$$

- $0 < y_i < 1$.
- $\sum_i y_i = 1$.
- $\mathbf{y} = \text{softmax}(\mathbf{z})$ is therefore a probability distribution over classes.
- The parameters $\mathbf{W}, \mathbf{b}$ are learned from the training data.

Numerically Stable Softmax

A useful implementation subtracts the largest logit before exponentiation:

$$\mathbf{z}' = \mathbf{z} - \max_i z_i.$$

Then

$$\text{softmax}(\mathbf{z})_i = \frac{e^{z'_i}}{\sum_c e^{z'_c}}.$$

Why?

Subtracting the same constant from all logits does not change the softmax probabilities, but reduces the risk of numerical overflow in e^{z_i} .

Learning Problem

Given training samples

$$\{(\mathbf{x}_n, \mathbf{t}_n)\}_{n=1}^N,$$

where $\mathbf{t}_n \in \{0, 1\}^K$ is one-hot, minimize the average categorical cross-entropy:

$$E(\mathbf{W}, \mathbf{b}) = \frac{1}{N} \sum_{n=1}^N E^n$$

with

$$E^n = - \sum_{k=1}^K t_k^n \log y_k^n.$$

Why Cross-Entropy?

- The model outputs probabilities, while the targets are one-hot categorical distributions.
- Cross-entropy measures the discrepancy between the target distribution and predicted distribution.
- For a one-hot target, only the probability assigned to the true class contributes directly to the loss.

$$E^n = -\log y_{\text{true}}^n.$$

Learning objective

Find **W** and **b** that minimize the average training loss.

Softmax Jacobian

Because softmax is vector-valued, its derivative with respect to $\mathbf{z}$ is a $K \times K$ Jacobian:

$$J_{ki} = \frac{\partial y_k}{\partial z_i}.$$

The two cases are

$$\frac{\partial y_k}{\partial z_i} = \begin{cases} y_i(1 - y_i), & k = i, \\ -y_i y_k, & k \neq i. \end{cases}$$

Equivalently,

$$\boxed{\frac{\partial y_k}{\partial z_i} = y_i(\delta_{ik} - y_k)}.$$

Softmax Jacobian in Matrix Form

$$\frac{\partial \mathbf{y}}{\partial \mathbf{z}} = \text{diag}(\mathbf{y}) - \mathbf{y}\mathbf{y}^T$$

- The diagonal contains the terms y_i .
- The outer product $\mathbf{y}\mathbf{y}^T$ contains all pairwise products.
- The Kronecker delta δ_{ik} is 1 when $i = k$ and 0 otherwise.

Gradient of Cross-Entropy with Respect to Weights

Applying the chain rule gives

$$\frac{\partial E}{\partial w_{ij}} = \frac{1}{N} \sum_{n=1}^N (y_i^n - t_i^n) x_j^n.$$

The important simplification is:

$$\boxed{\frac{\partial E}{\partial w_{ij}} = \frac{1}{N} \sum_n (y_i^n - t_i^n) x_j^n.}$$

- The softmax Jacobian and cross-entropy derivative simplify together.
- The error term is simply predicted probability minus target.

Gradient with Respect to Bias

For the bias parameter:

$$\frac{\partial E}{\partial b_i} = \frac{1}{N} \sum_{n=1}^N (y_i^n - t_i^n).$$

Thus the same probability error drives both parameter gradients:

$$y_i^n - t_i^n.$$

Interpretation

If a class probability is too high, its parameters receive a positive gradient; if it is too low, the gradient has the opposite sign.

Gradient-Descent Updates

The update equations are

$$w_{ij}^{t+1} = w_{ij}^t - \eta \frac{1}{N} \sum_{n=1}^N (y_i^n - t_i^n) x_j^n,$$

$$b_i^{t+1} = b_i^t - \eta \frac{1}{N} \sum_{n=1}^N (y_i^n - t_i^n).$$

- η is the learning rate.
- Full-batch gradient descent uses all N samples per update.
- Mini-batch stochastic gradient descent can make training more efficient for large datasets.

Vectorized Gradient Form

For a batch represented by matrix X :

$$Z = XW + b, \quad Y = \text{softmax}(Z).$$

The gradients can be written compactly as

$$\nabla_w E = \frac{1}{N} X^T (Y - T)$$

$$\nabla_b E = \frac{1}{N} \sum_{n=1}^N (Y_n - T_n).$$

This is the form used naturally in NumPy/PyTorch implementations.

Classification Decision

After learning $\mathbf{W}$ and $\mathbf{b}$, prediction is made by selecting the class with the largest probability:

$$\hat{y} = \arg \max_{1 \leq k \leq K} y_k$$

Because softmax is monotonic with respect to each individual logit,

$$\hat{y} = \arg \max_{1 \leq k \leq K} z_k .$$

Thus prediction can be made directly from the logits.

Linear Decision Boundaries

For three classes and two input variables:

$$z_i = w_{i1}x_1 + w_{i2}x_2 + b_i.$$

The boundary between classes 1 and 2 occurs where their logits are equal:

$$z_1 = z_2.$$

Therefore,

$$(w_{11} - w_{21})x_1 + (w_{12} - w_{22})x_2 = b_2 - b_1.$$

This is a straight line in the two-dimensional feature space.

Three-Class Decision Boundaries

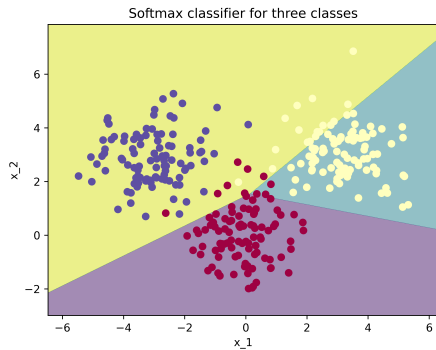


Figure: Linear decision boundaries between three classes.

Three-Class Example: Data Generation

The example generates three Gaussian clusters:

- Class 0: mean $[0, 0]$
- Class 1: mean $[3, 3]$
- Class 2: mean $[-3, 3]$
- 100 samples per class
- One-hot targets for the three classes

$$X = \text{concatenate}(X_0, X_1, X_2), \quad T = \text{concatenate}(T_0, T_1, T_2).$$

Python: Softmax and Cross-Entropy

```
def softmax(x):  
    x = x - np.max(x, axis=1, keepdims=True)  
    exp_x = np.exp(x)  
    return exp_x / np.sum(exp_x, axis=1, keepdims=True)  
  
def cross_entropy(y_true, y_pred):  
    y_pred = np.clip(y_pred, 1e-12, 1 - 1e-12)  
    return -np.mean(np.sum(y_true * np.log(y_pred), axis=1))
```

Python: Gradient Descent

```
def gradient_descent(X, y_true, y_pred, w, b, learning_rate):  
    m = X.shape[0]  
    dw = (1/m) * np.dot(X.T, (y_pred - y_true))  
    db = (1/m) * np.sum(y_pred - y_true, axis=0)  
    w = w - learning_rate * dw  
    b = b - learning_rate * db  
    return w, b
```

Core computation

The implementation is exactly the vectorized form of the derived gradients.

Python: Training Loop

```
w = np.random.randn(2, 3)
b = np.random.randn(3)
epochs = 100
learning_rate = 0.01

for epoch in range(epochs):
    y_pred = softmax(np.dot(X, w) + b)
    loss = cross_entropy(y_true, y_pred)
    if epoch % 10 == 0:
        print(f"Epoch {epoch}, Loss: {loss:.4f}")
    w, b = gradient_descent(
        X, y_true, y_pred, w, b, learning_rate)
```

Python: Decision Regions

```
xx, yy = np.meshgrid(
    np.arange(x_min, x_max, .01),
    np.arange(y_min, y_max, .01))

Z = np.argmax(
    softmax(np.c_[xx.ravel(), yy.ravel()] @ w + b),
    axis=1).reshape(xx.shape)

plt.contourf(xx, yy, Z, alpha=.5)
plt.scatter(X[:,0], X[:,1],
            c=np.argmax(y_true, axis=1))
plt.show()
```

Example: What the Training Learns

- The model starts with randomly initialized weights and biases.
- Each iteration computes class probabilities with softmax.
- Cross-entropy measures the prediction error.
- The gradients adjust W and b toward lower loss.
- The resulting model creates linear regions separated by linear decision boundaries.

Key point

For this single-layer softmax model, the learned decision boundaries are linear in the input feature space.

Summary

- 1 Softmax regression extends linear classification to K classes.
- 2 Class labels are represented as one-hot vectors.
- 3 Softmax converts logits into a probability distribution.
- 4 Categorical cross-entropy provides the learning objective.
- 5 The softmax Jacobian leads to a simple $Y - T$ gradient.
- 6 Gradient descent learns W and b .
- 7 Prediction selects the largest logit/probability.
- 8 The single-layer classifier produces linear decision boundaries.

Assignment

Handwritten-Digit Recognition

Implement optical recognition of handwritten digits as a multiclass classification task using Python from scratch, using the Keras deep-learning library, and Google Colab.

- Load the Optical Recognition of Handwritten Digits dataset.
- Encode the digit labels for multiclass classification.
- Train a suitable classifier.
- Evaluate the classification performance.
- Visualize or analyze the predictions.

Dataset source:

<https://archive.ics.uci.edu/ml/datasets/Optical+Recognition+of+Handwritten+Digits>

Thank You

Questions?